

AWS Well Architected Framework Review Report

WAFPillar	Reliability
Question	How do you plan for disaster recovery (DR)?
BestPractice	Define recovery objectives for downtime and data loss
Severity	High
Reference	https://aws.amazon.com/
Validation	RTO/RPO recovery objectives
Status	Requires manual review - verify RTO/RPO definition

WAFPillar	Reliability
Question	How do you plan for disaster recovery (DR)?
BestPractice	Use defined recovery strategies to meet the recovery objectives
Severity	High
Reference	https://aws.amazon.com/
Validation	Backup plans for recovery strategy
Status	Backup plans: 0

WAFPillar	Reliability
Question	How do you plan for disaster recovery (DR)?
BestPractice	Test disaster recovery implementation to validate the implementation
Severity	High
Reference	https://aws.amazon.com/

Validation	DR implementation testing
Status	Requires manual review - verify DR testing

WAFPillar	Reliability
Question	How do you plan for disaster recovery (DR)?
BestPractice	Manage configuration drift at the DR site or Region
Severity	High
Reference	https://aws.amazon.com/
Validation	CloudFormation stack drift
Status	Stacks with drift info: 12, drifted: 0

WAFPillar	Reliability
Question	How do you plan for disaster recovery (DR)?
BestPractice	Automate recovery
Severity	High
Reference	https://aws.amazon.com/
Validation	Automated recovery mechanisms
Status	Backup plans: 0, Route53 health checks: 4

WAFPillar	Reliability
Question	How do you test reliability?
BestPractice	Use playbooks to investigate failures
Severity	High

Reference	https://aws.amazon.com/
Validation	SSM automation documents for failure investigation
Status	10 SSM automation documents found

WAF Pillar	Reliability
Question	How do you test reliability?
Best Practice	Perform post-incident analysis
Severity	High
Reference	https://aws.amazon.com/
Validation	Post-incident analysis process
Status	Requires manual review

WAF Pillar	Reliability
Question	How do you test reliability?
Best Practice	Test functional requirements
Severity	High
Reference	https://aws.amazon.com/
Validation	CI/CD pipelines for functional testing
Status	No CodePipeline pipelines found

WAF Pillar	Reliability
Question	How do you test reliability?
Best Practice	Test scaling and performance requirements

Severity	High
Reference	https://aws.amazon.com/
Validation	Scaling and performance testing
Status	Requires manual review

WAFPillar	Reliability
Question	How do you test reliability?
BestPractice	Test resiliency using chaos engineering
Severity	Medium
Reference	https://aws.amazon.com/
Validation	AWS FIS experiments for chaos engineering
Status	No FIS experiments found - consider chaos engineering

WAFPillar	Reliability
Question	How do you test reliability?
BestPractice	Conduct game days regularly
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Game day exercises
Status	Requires manual review

WAFPillar	Reliability
Question	How do you design your workload to withstand component failures?

BestPractice	Monitor all components of the workload to detect failures
Severity	High
Reference	https://aws.amazon.com/
Validation	CloudWatch alarms for failure detection
Status	CloudWatch alarms: 0

WAFPillar	Reliability
Question	How do you design your workload to withstand component failures?
BestPractice	Fail over to healthy resources
Severity	High
Reference	https://aws.amazon.com/
Validation	Healthy targets in target groups
Status	Healthy targets: 0/0

WAFPillar	Reliability
Question	How do you design your workload to withstand component failures?
BestPractice	Automate healing on all layer
Severity	High
Reference	https://aws.amazon.com/
Validation	ASG health checks and replacement
Status	No ASGs

WAFPillar	Reliability
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Question	How do you design your workload to withstand component failures?
BestPractice	Rely on the data plane and not the control plane during recover
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Data plane vs control plane during recovery
Status	Requires manual review - verify data plane recovery

WAFPillar	Reliability
Question	How do you design your workload to withstand component failures?
BestPractice	Use static stability to prevent bimodal behavior
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Static stability design
Status	Requires manual review - verify static stability

WAFPillar	Reliability
Question	How do you design your workload to withstand component failures?
BestPractice	Send notifications when events impact availability
Severity	Medium
Reference	https://aws.amazon.com/
Validation	SNS notification topics
Status	SNS topics: 3

WAFPillar	Reliability
Question	How do you design your workload to withstand component failures?
BestPractice	Architect your product to meet availability targets and uptime service level agreements
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Availability targets and SLAs
Status	Requires manual review - verify availability targets

WAFPillar	Reliability
Question	How do you use fault isolation to protect your workload?
BestPractice	Deploy the workload to multiple locations
Severity	High
Reference	https://aws.amazon.com/
Validation	EC2 instances across AZs
Status	Running instances: 0 across 0 AZs

WAFPillar	Reliability
Question	How do you use fault isolation to protect your workload?
BestPractice	Select the appropriate locations for your multi-location deployment
Severity	High
Reference	https://aws.amazon.com/
Validation	RDS Multi-AZ deployment
Status	RDS Multi-AZ: 0, single-AZ: 0

WAFPillar	Reliability
Question	How do you use fault isolation to protect your workload?
BestPractice	Automate recovery for components constrained to a single location
Severity	Medium
Reference	https://aws.amazon.com/
Validation	ASG multi-AZ spanning
Status	ASGs spanning multiple AZs: 0

WAFPillar	Reliability
Question	How do you use fault isolation to protect your workload?
BestPractice	Use bulkhead architectures to limit scope of impact
Severity	High
Reference	https://aws.amazon.com/
Validation	VPC/subnet isolation (bulkhead)
Status	VPCs: 1, Subnets: 6 (isolation)

WAFPillar	Reliability
Question	How do you back up data
BestPractice	Identify and back up all data that needs to be backed up, or reproduce the data from sources
Severity	High
Reference	https://aws.amazon.com/
Validation	AWS Backup plans

Status	Backup plans: 0
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WAFPillar	Reliability
Question	How do you back up data
BestPractice	Secure and encrypt backups
Severity	High
Reference	https://aws.amazon.com/
Validation	Backup vault encryption and access policy
Status	Backup vaults: 0, encrypted: 0

WAFPillar	Reliability
Question	How do you back up data
BestPractice	Perform data backup automatically
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Automated backup schedule
Status	Plans with schedule: 0

WAFPillar	Reliability
Question	How do you back up data
BestPractice	Perform periodic recovery of the data to verify backup integrity and processes
Severity	Medium
Reference	https://aws.amazon.com/

Validation	Periodic recovery verification
Status	Requires manual review - verify recovery testing

WAFPillar	Reliability
Question	How do you implement change
BestPractice	Use runbooks for standard activities such as deployment
Severity	High
Reference	https://aws.amazon.com/
Validation	SSM Automation runbooks
Status	SSM Automation documents: 10

WAFPillar	Reliability
Question	How do you implement change
BestPractice	Integrate functional testing as part of your deployment
Severity	High
Reference	https://aws.amazon.com/
Validation	CodePipeline for functional testing
Status	CodePipeline pipelines: 0

WAFPillar	Reliability
Question	How do you implement change
BestPractice	Integrate resiliency testing as part of your deployment
Severity	Medium

Reference	https://aws.amazon.com/
Validation	Resiliency testing in deployment
Status	Requires manual review - verify resiliency testing

WAFPillar	Reliability
Question	How do you implement change
BestPractice	Deploy using immutable infrastructure
Severity	Medium
Reference	https://aws.amazon.com/
Validation	CloudFormation for immutable infrastructure
Status	CloudFormation stacks: 12

WAFPillar	Reliability
Question	How do you implement change
BestPractice	Deploy changes with automation
Severity	Medium
Reference	https://aws.amazon.com/
Validation	CodeDeploy for automated deployment
Status	CodeDeploy applications: 0

WAFPillar	Reliability
Question	How do you design your workload to adapt to changes in demand?
BestPractice	Use automation when obtaining or scaling resources

Severity	Medium
Reference	https://aws.amazon.com/
Validation	Auto Scaling groups
Status	Auto Scaling groups: 0

WAFPillar	Reliability
Question	How do you design your workload to adapt to changes in demand?
BestPractice	Obtain resources upon detection of impairment to a workload
Severity	Medium
Reference	https://aws.amazon.com/
Validation	ASG health check for impairment detection
Status	No ASGs

WAFPillar	Reliability
Question	How do you design your workload to adapt to changes in demand?
BestPractice	Obtain resources upon detection that more resources are needed for a workload
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Scaling policies
Status	Scaling policies: 0

WAFPillar	Reliability
Question	How do you design your workload to adapt to changes in demand?

BestPractice	Load test your workload
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Load testing
Status	Requires manual review - verify load testing

WAFPillar	Reliability
Question	How do you monitor workload resources?
BestPractice	Monitor all components for the workload (Generation)
Severity	High
Reference	https://aws.amazon.com/
Validation	CloudWatch alarms for workload components
Status	CloudWatch alarms: 0

WAFPillar	Reliability
Question	How do you monitor workload resources?
BestPractice	Define and calculate metrics (Aggregation)
Severity	High
Reference	https://aws.amazon.com/
Validation	CloudWatch dashboards for metrics
Status	CloudWatch dashboards: 0

WAFPillar	Reliability
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Question	How do you monitor workload resources?
BestPractice	Send notifications (Real-time processing and alarming)
Severity	High
Reference	https://aws.amazon.com/
Validation	SNS subscriptions for notifications
Status	SNS subscriptions: 0

WAFPillar	Reliability
Question	How do you monitor workload resources?
BestPractice	Automate responses (Real-time processing and alarming)
Severity	Medium
Reference	https://aws.amazon.com/
Validation	EventBridge rules for automation
Status	EventBridge rules: 2

WAFPillar	Reliability
Question	How do you monitor workload resources?
BestPractice	Analyze logs
Severity	Medium
Reference	https://aws.amazon.com/
Validation	CloudWatch Logs groups
Status	Log groups: 10

WAFPillar	Reliability
Question	How do you monitor workload resources?
BestPractice	Conduct reviews regularly
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Regular reviews
Status	Requires manual review - verify regular review cadence

WAFPillar	Reliability
Question	How do you monitor workload resources?
BestPractice	Monitor end-to-end tracing of requests through your system
Severity	Medium
Reference	https://aws.amazon.com/
Validation	X-Ray tracing
Status	X-Ray sampling rules: 1 (tracing configured)

WAFPillar	Reliability
Question	How do you design interactions in a distributed system to mitigate or withstand failures
BestPractice	Implement graceful degradation to transform applicable hard dependencies into soft dependencies
Severity	High
Reference	https://aws.amazon.com/
Validation	Graceful degradation design
Status	Requires manual review - verify graceful degradation

WAFPillar	Reliability
Question	How do you design interactions in a distributed system to mitigate or withstand failures
BestPractice	Throttle requests
Severity	High
Reference	https://aws.amazon.com/
Validation	API Gateway throttle settings
Status	Usage plans with throttle: 0

WAFPillar	Reliability
Question	How do you design interactions in a distributed system to mitigate or withstand failures
BestPractice	Control and limit retry calls
Severity	High
Reference	https://aws.amazon.com/
Validation	Retry control and limits
Status	Requires manual review - verify retry limits

WAFPillar	Reliability
Question	How do you design interactions in a distributed system to mitigate or withstand failures
BestPractice	Fail fast and limit queues
Severity	High
Reference	https://aws.amazon.com/
Validation	Dead letter queues for fail-fast

Status	Queues with DLQ: 0
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WAFPillar	Reliability
Question	How do you design interactions in a distributed system to mitigate or withstand failures
BestPractice	Set client timeouts
Severity	High
Reference	https://aws.amazon.com/
Validation	Client timeouts
Status	Requires manual review - verify client timeouts

WAFPillar	Reliability
Question	How do you design interactions in a distributed system to mitigate or withstand failures
BestPractice	Make systems stateless where possible
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Stateless design via ElastiCache
Status	ElastiCache clusters: 0 (stateless sessions)

WAFPillar	Reliability
Question	How do you design interactions in a distributed system to mitigate or withstand failures
BestPractice	Implement emergency levers
Severity	Medium
Reference	https://aws.amazon.com/

Validation	Emergency levers
Status	Requires manual review - verify emergency levers

WAFPillar	Reliability
Question	How do you design interactions in a distributed system to prevent failures?
BestPractice	Identify the kind of distributed systems you depend on
Severity	High
Reference	https://aws.amazon.com/
Validation	Distributed system components (SQS, SNS)
Status	Async components - SQS queues: 1, SNS topics: 3

WAFPillar	Reliability
Question	How do you design interactions in a distributed system to prevent failures?
BestPractice	Implement loosely coupled dependencies
Severity	High
Reference	https://aws.amazon.com/
Validation	Loosely coupled dependencies (queues)
Status	SQS queues (loose coupling): 1

WAFPillar	Reliability
Question	How do you design interactions in a distributed system to prevent failures?
BestPractice	Do constant work
Severity	Low

Reference	https://aws.amazon.com/
Validation	Constant work pattern
Status	Requires manual review - verify constant work pattern

WAFPillar	Reliability
Question	How do you design interactions in a distributed system to prevent failures?
BestPractice	Make all responses idempotent
Severity	Low
Reference	https://aws.amazon.com/
Validation	Idempotent responses
Status	Requires manual review - verify idempotent design

WAFPillar	Reliability
Question	How do you design your workload service architecture?
BestPractice	Choose how to segment your workload
Severity	High
Reference	https://aws.amazon.com/
Validation	Workload segmentation via VPCs and subnets
Status	VPCs: 1, Subnets: 6 (segmentation)

WAFPillar	Reliability
Question	How do you design your workload service architecture?
BestPractice	Build services focused on specific business domains and functionality

Severity	High
Reference	https://aws.amazon.com/
Validation	Business-domain-focused services
Status	Requires manual review - verify service domain alignment

WAFPillar	Reliability
Question	How do you design your workload service architecture?
BestPractice	Provide service contracts per API
Severity	Medium
Reference	https://aws.amazon.com/
Validation	API service contracts
Status	REST APIs: 0

WAFPillar	Reliability
Question	How do you plan your network topology?
BestPractice	Use highly available network connectivity for your workload public endpoints
Severity	High
Reference	https://aws.amazon.com/
Validation	ALB availability across AZs
Status	ALBs: 0, across 0 AZs

WAFPillar	Reliability
Question	How do you plan your network topology?

BestPractice	Provision redundant connectivity between private networks in the cloud and on-premises environments
Severity	High
Reference	https://aws.amazon.com/
Validation	Redundant connectivity (Direct Connect/VPN)
Status	Direct Connect: 0, VPN connections: 0

WAFPillar	Reliability
Question	How do you plan your network topology?
BestPractice	Ensure IP subnet allocation accounts for expansion and availability
Severity	High
Reference	https://aws.amazon.com/
Validation	Subnet allocation for expansion and availability
Status	Subnets: 6, across 6 AZs

WAFPillar	Reliability
Question	How do you plan your network topology?
BestPractice	Prefer hub-and-spoke topologies over many-to-many mesh
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Hub-and-spoke via Transit Gateway
Status	No Transit Gateways found

WAFPillar	Reliability
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Question	How do you plan your network topology?
BestPractice	Enforce non-overlapping private IP address ranges in all private address spaces where they are connected
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Non-overlapping VPC CIDR blocks
Status	VPCs checked: 1 CIDR blocks, no overlaps detected

WAFPillar	Reliability
Question	How do you manage service quotas and constraints?
BestPractice	Aware of service quotas and constraint
Severity	High
Reference	https://aws.amazon.com/
Validation	EC2 service quotas awareness
Status	EC2 service quotas: 18 quotas listed

WAFPillar	Reliability
Question	How do you manage service quotas and constraints?
BestPractice	Manage service quotas across accounts and Regions
Severity	High
Reference	https://aws.amazon.com/
Validation	Multi-account and cross-region quota management
Status	Requires manual review - verify quota management across accounts/Regions

WAFPillar	Reliability
Question	How do you manage service quotas and constraints?
BestPractice	Accommodate fixed service quotas and constraints through architecture
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Architecture accommodating fixed quotas
Status	Requires manual review - verify architecture accounts for quota constraints

WAFPillar	Reliability
Question	How do you manage service quotas and constraints?
BestPractice	Monitor and manage quotas
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Quota-related CloudWatch alarms
Status	Quota-related alarms found: 0

WAFPillar	Reliability
Question	How do you manage service quotas and constraints?
BestPractice	Automate quota management
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Automated quota management
Status	Requires manual review - verify quota automation

WAFPillar	Reliability
Question	How do you manage service quotas and constraints?
BestPractice	Ensure that a sufficient gap exists between the current quotas and the maximum usage to accommodate failover
Severity	Medium
Reference	https://aws.amazon.com/
Validation	Quota headroom for failover
Status	Requires manual review - verify quota gap for failover